



Embedded Software for the Internet Of Things

The Internet of Things

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The Internet of Things (IoT)

- Internet of Things (IoT) refers to a **large ecosystem** of devices and smart objects:
 - that can gather, exchange, and act on information over **networking infrastructure**.



The Internet of Things (IoT)

- These “things” include:
 - Home appliances, e.g., washing machines, fridges, kettles, vacuum cleaners, light bulbs, thermostats, door locks, and video cameras
 - Civil engineering structures, e.g., bridges and railways
 - Wearable devices, e.g., smart watches, smart glasses, rings, and clothes
 - Entertainment devices, e.g., TVs, game consoles, and toys
 - Biomedical devices, e.g., blood pressure monitors, digital pills
- **Any THING** in the world!



Objects that make up the IoT



They incorporate computing hardware with small form-factor



Have (multiple) sensing and/or actuation capabilities



Often have low-power consumption and can be battery-powered



Usually transmit and receive information wirelessly



They are connected to the Internet and may be uniquely addressable



Their functionality can be (re)programmed



May be able to operate autonomously without human interaction

From embedded systems to IoT

- IoT can be seen as an evolution of wireless sensor networks, where the network is no longer dedicated to a single application.
- Instead IoT provides a platform that can accommodate a range of applications and devices that are easier to manage.
- Like in the ubiquitous computing paradigm, IoT devices operate seamlessly, but in addition
 - they are connected to the Internet,
 - have more functionality, and
 - are more intelligent.

Technologies that enable IoT



Mobile computing:

Faster, cheaper, low-power processors and devices (e.g., smartphones and tablets)



Little devices and more sensing options:

Wearables and rich data sources



Big data and the cloud:

Cloud-computing services to process data from sensors

Technologies that enable IoT



Better batteries:

E.g., lithium-ion polymer – more energy storage, rapid charging



Faster and more diverse standardized connectivity options:

Short-range (Bluetooth, Zigbee), medium-range (Wi-Fi, cellular), and long-range (LoRa, SigFox)

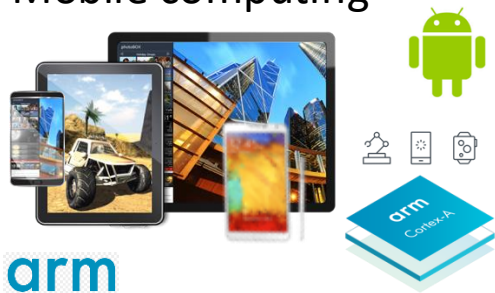


Development resources:

APIs and protocols (REST, MQTT), language-independent data formats (e.g., JSON), development communities (Mbed)

Technologies that enable IoT

Mobile computing



arm
CORTEX®

Little devices and more sensors/Big data
and the cloud



Devices



Applications



Big Data



arm MBED



1,000,000s



Platform

Cloud and community-based
development

IEEE 802.15.4



WiFi™



LTE

Bluetooth®
SMART



ZigBee®

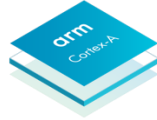
Networking standards that enable scalability

Technologies that enable IoT

Mobile computing



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Technologies that enable IoT

Little devices and more sensors/Big data
and the cloud



Devices



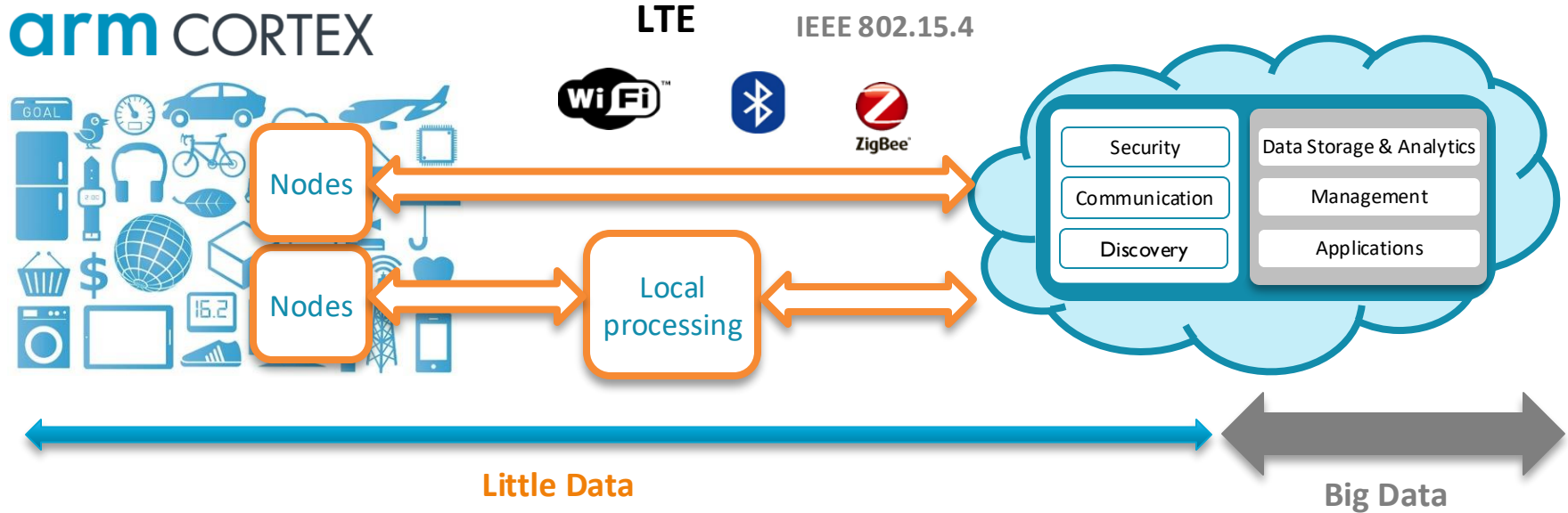
Applications



Big Data



Little devices and sensing (little data)



Technologies that enable IoT



Networking standards that enable scalability

Bluetooth Low Energy (BLE) connectivity

- **Ultra low power**, designed to run with coin cells, e.g., 10mW
 - Short range communication, e.g., < 100 meters
 - Powering small IoT devices, e.g., wearables
 - Proximity sensing/inventory tracking (BLE beacons)
- Many-to-many communications (mesh networking), e.g. building automation



Wi-Fi connectivity



- Longer communication range
- Uses unlicensed spectrum and simple decentralized channel access protocol
→ easy to deploy
- Higher throughput supported
- Particularly suitable for devices with less stringent power constraints, e.g., smart home appliances
 - Power-saving mode (PSM) also defined to save energy in battery-powered devices

Technologies that enable IoT



arm MBED



1,000,000s



Platform

Cloud and community-based
development

Challenges facing IoT

- Interconnecting many devices that exchange (big) data can be challenging



Better hardware
and code that
runs efficiently.



Ensuring energy
efficient
operation.



Aggregating large
data sets and
acting on it in
real-time.



Modeling and
predicting the
behavior of
complex systems.



Ensuring reliable
connectivity and
scalability.



Guaranteeing
secure operation
and data
confidentiality



Developing easy
to use products
and services



Complying with
regulations

Example: Wearable specific constraints

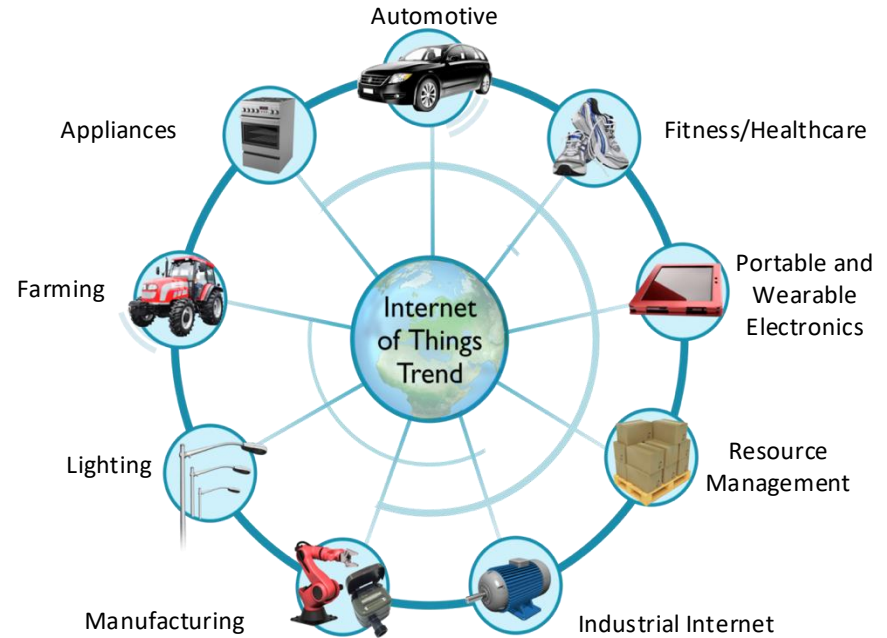
- Wristband fitness trackers monitoring activity, sleep, heart rate, etc.
 - Limited memory
 - Battery powered
 - Minimal user interface
 - Short range communication
 - *Bluetooth Low Energy (BLE)*
 - Wireless communication
 - *Risks of eavesdropping, surveillance, etc.*



Opportunities

- **Application domains**

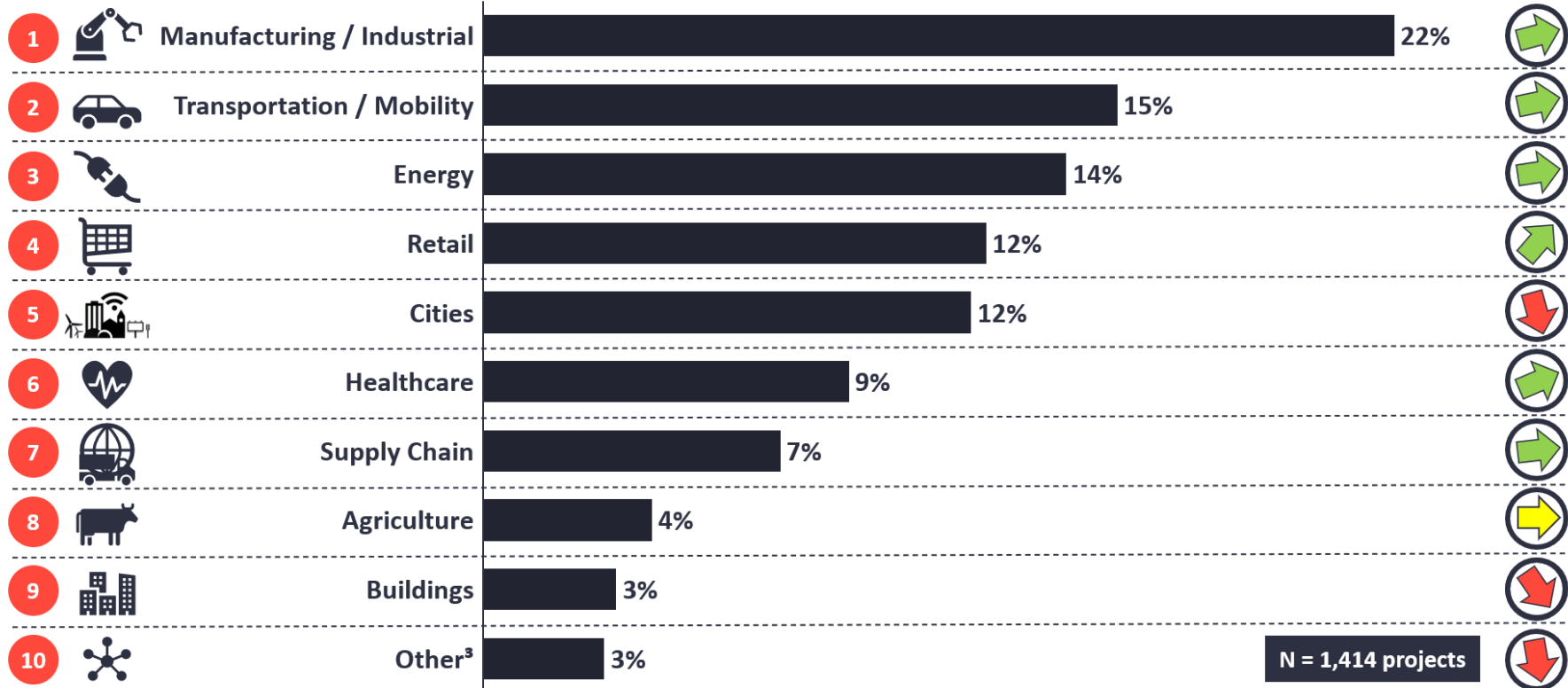
- Industry (manufacturing, transportation, agriculture)
- Consumer (smart homes, appliances, assisted living)
- Wearables (healthcare, fitness, productivity)



Top 10 IoT Application areas 2020

Global share of Enterprise IoT projects¹

Trend²

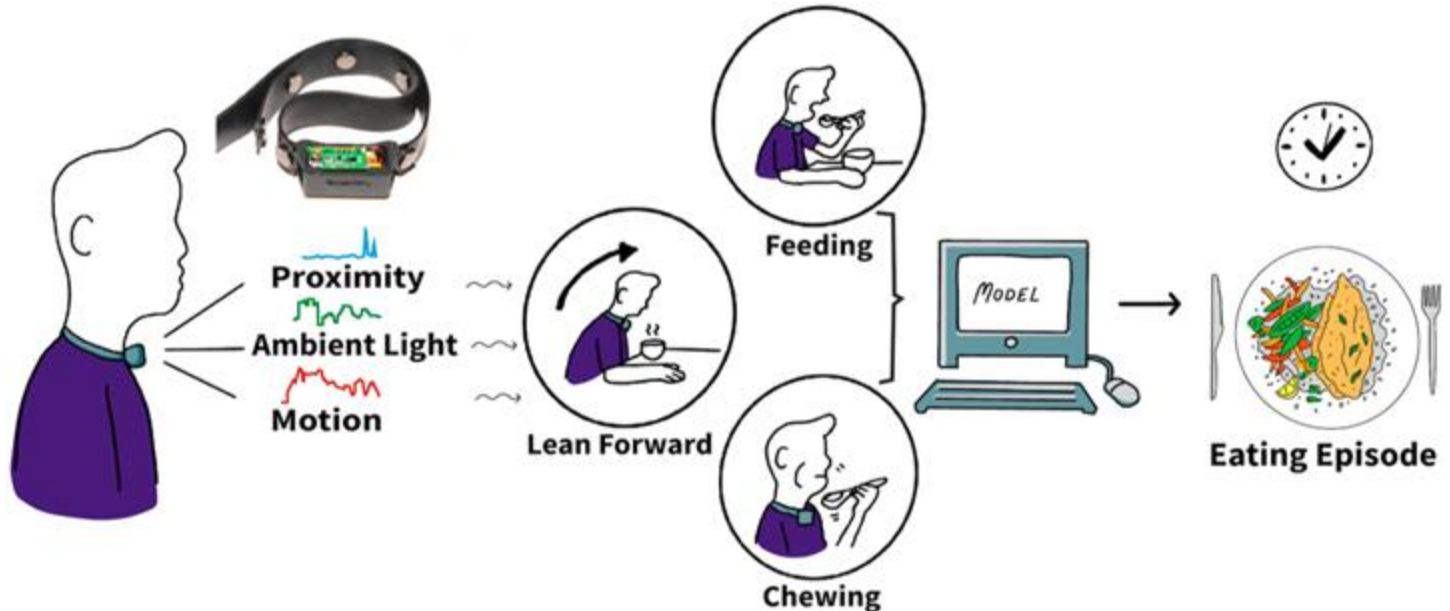


N = 1,414 projects

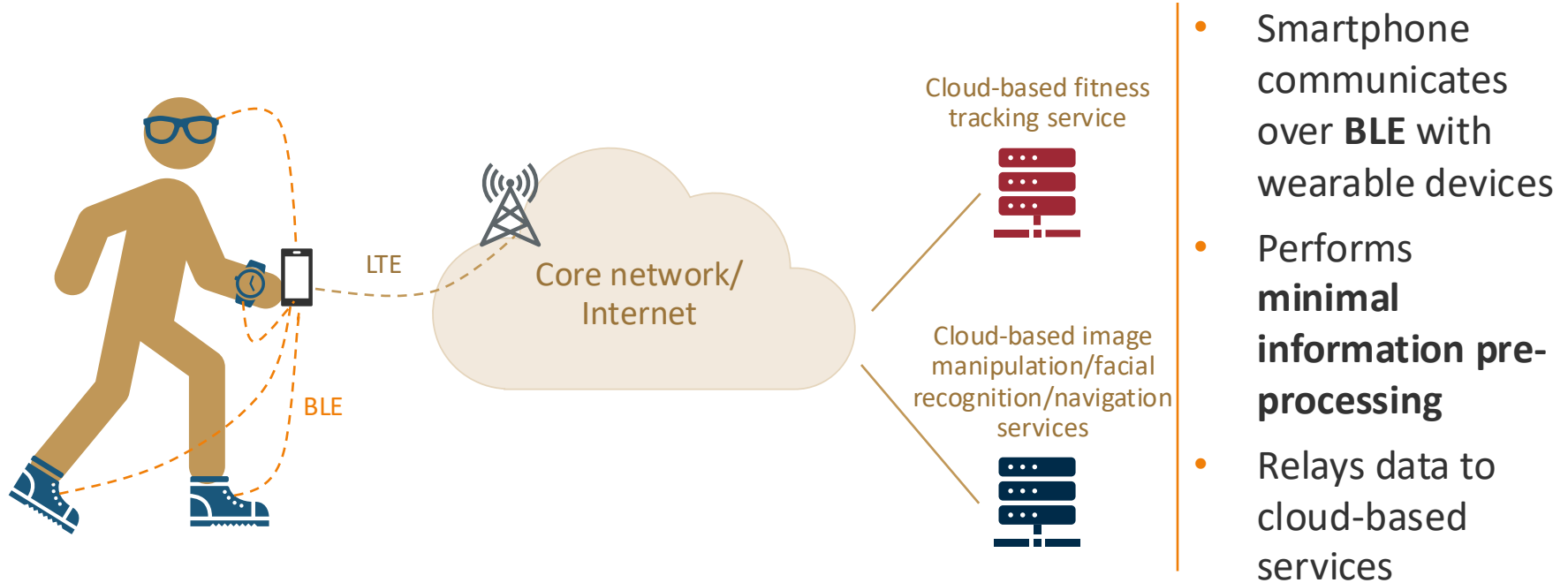
Note: 1. Based on 1,414 publicly known IoT projects (not including consumer IoT projects eg smart home, wearables, etc.) 2. Trend based on relative comparison with % of projects in the 2018 IoT Analytics IoT project list e.g., a downward arrow means the relative share of all projects has declined, not the overall number of projects. 3. Other includes IoT projects from Enterprise & Finance sectors. **Source:** IoT Analytics Research - July 2020

Healthcare Applications

- Implanted devices in the body.
- Wearable devices on the body.
- Environmental devices outside the body.



Wearables

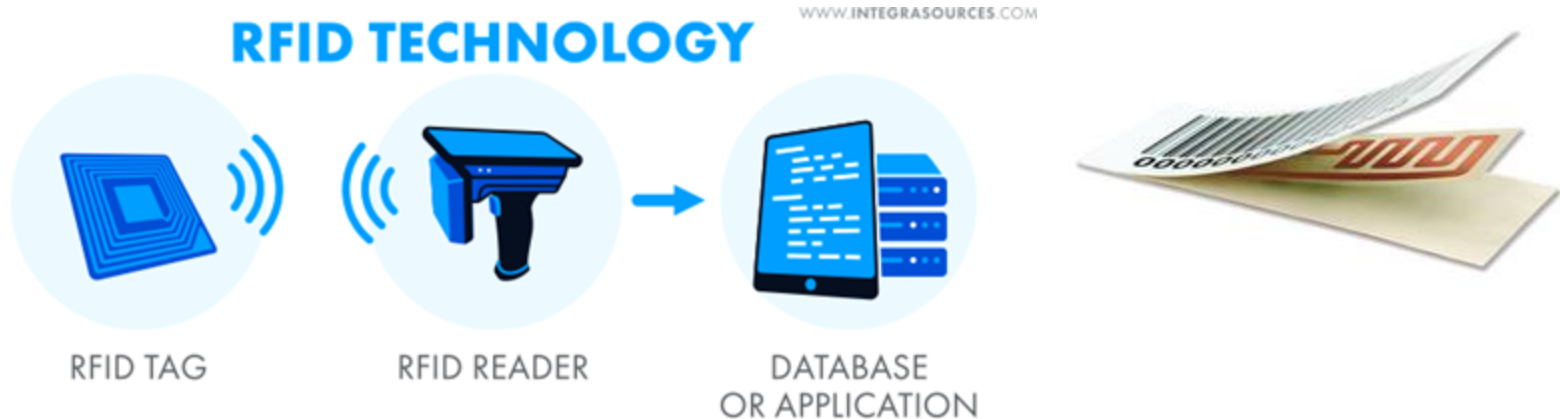


Smart Home



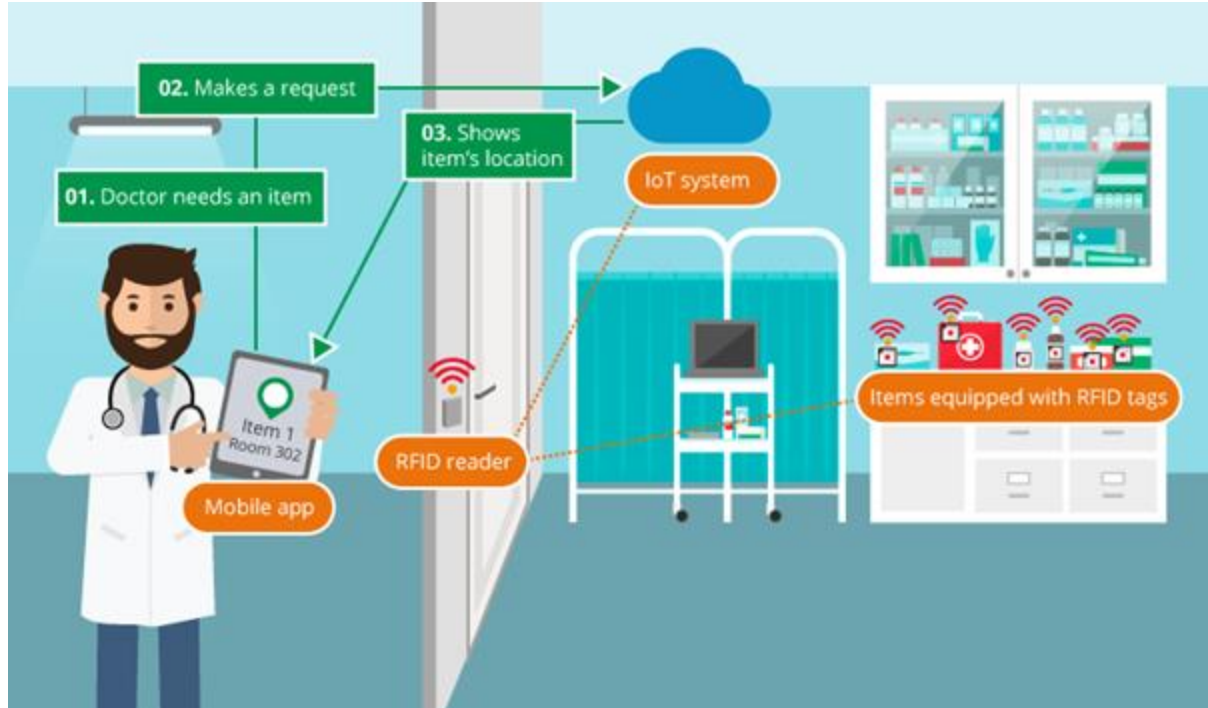
RFID Applications

- RFID tag can provide object ID (Electronic Product Code, etc.)
 - Many tags are read-only, some are writable.
- Two types of tags:
 - Passive transmits only when it receives a request.
 - Active tag both transmits independently and responds to requests.



RFID Applications

- RFID tags may operate in several different bands and at different ranges.



Smart Farms



Smart Agriculture



(a)



(b)



(c)

Smart trap

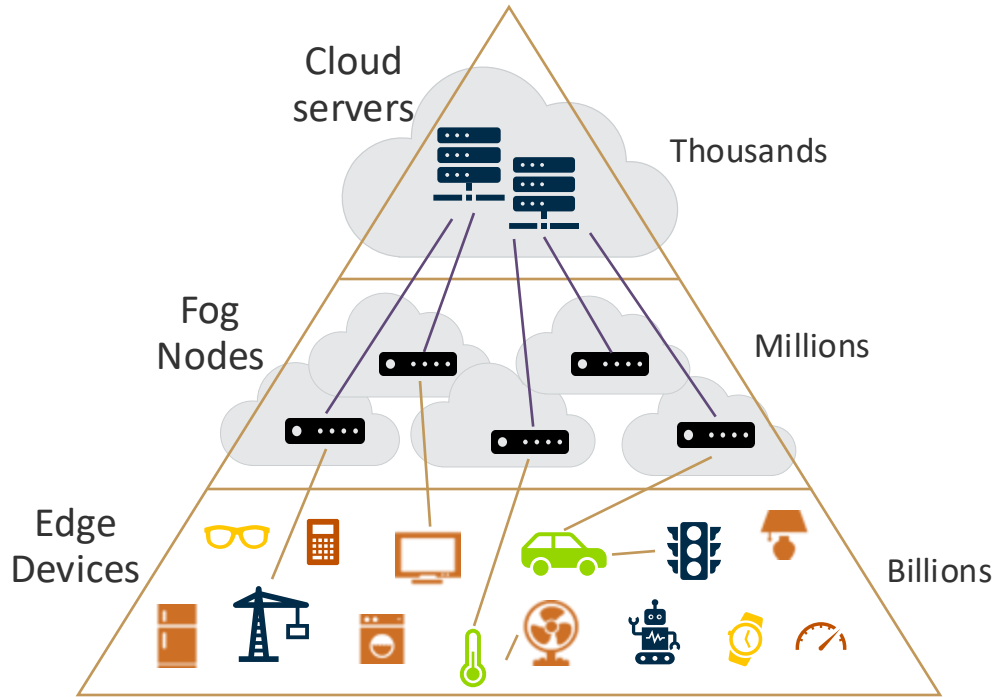
Smart reporting and reliable forecasting on pest situation



Sticky papers with captured moths

The **codling moth** (*Cydia pomonella*) is the most harmful pest in apple and pear orchards.

IoT architecture



- What are the application requirements?
- What data needs to be acted on locally?
- Where is most of the computing power?
- How much networking infrastructure should be deployed/used?

Cloud vs. Fog vs. Edge



Cloud computing



Cloud dominated the networked systems landscape until recently



All intelligence on powerful servers, including relational databases, control functions, data analytics engines, web interfaces, etc.

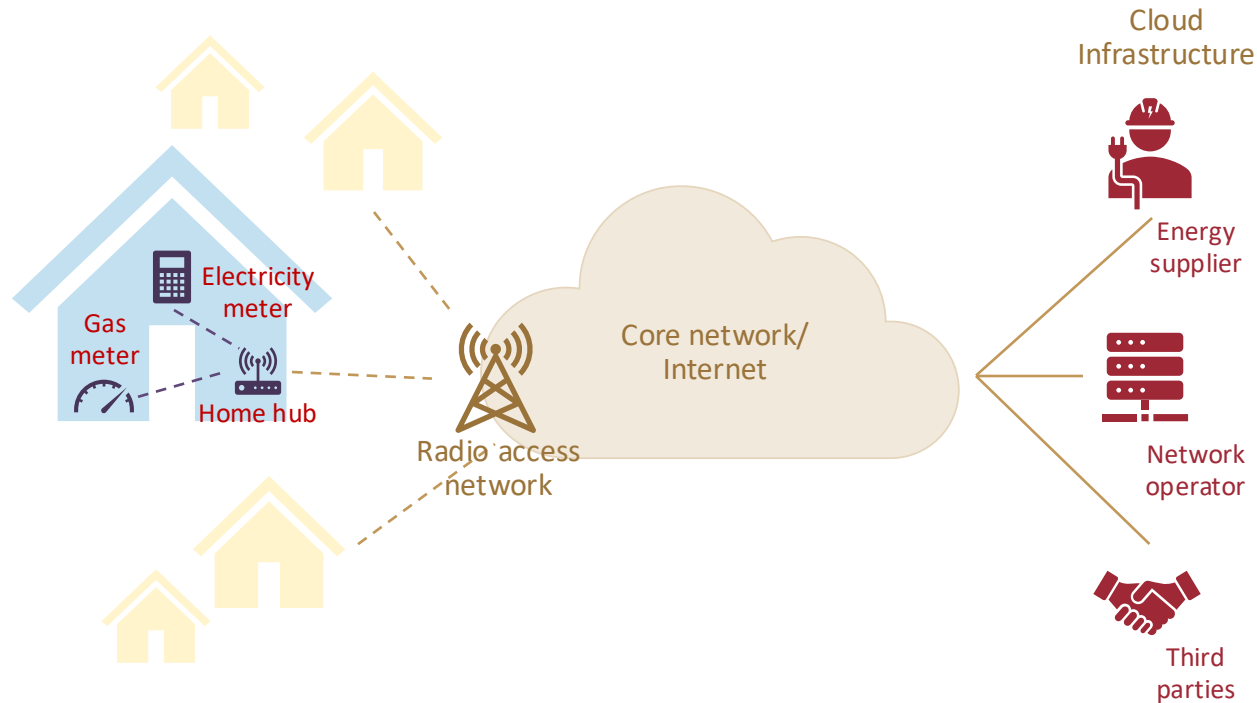


End devices merely information gatherers



Might not scale as the number of IoT devices grows, and applications continue to diversify and generate more data

Smart metering



- Sensing performed by **simple sensors**
- Information relayed by home hub over **cellular network**
- Data processed in the cloud by different stakeholders

Cloud vs. Fog vs. Edge



Pushing some of the intelligence closer to the device, for e.g., to access networks or gateways



This includes data aggregation, compression, (partial) processing, making localized decisions



IoT devices kept simple, no direct communication with end servers, still battery powered



Resource management implemented across different network layers – management could be regarded as an application



Fog computing

The role of gateways in fog architectures



Data filtering and processing (for e.g., aggregation of summaries, compression, etc.)



Protocol translation and interfacing between different connectivity technologies



Data flow multiplexing, packet routing



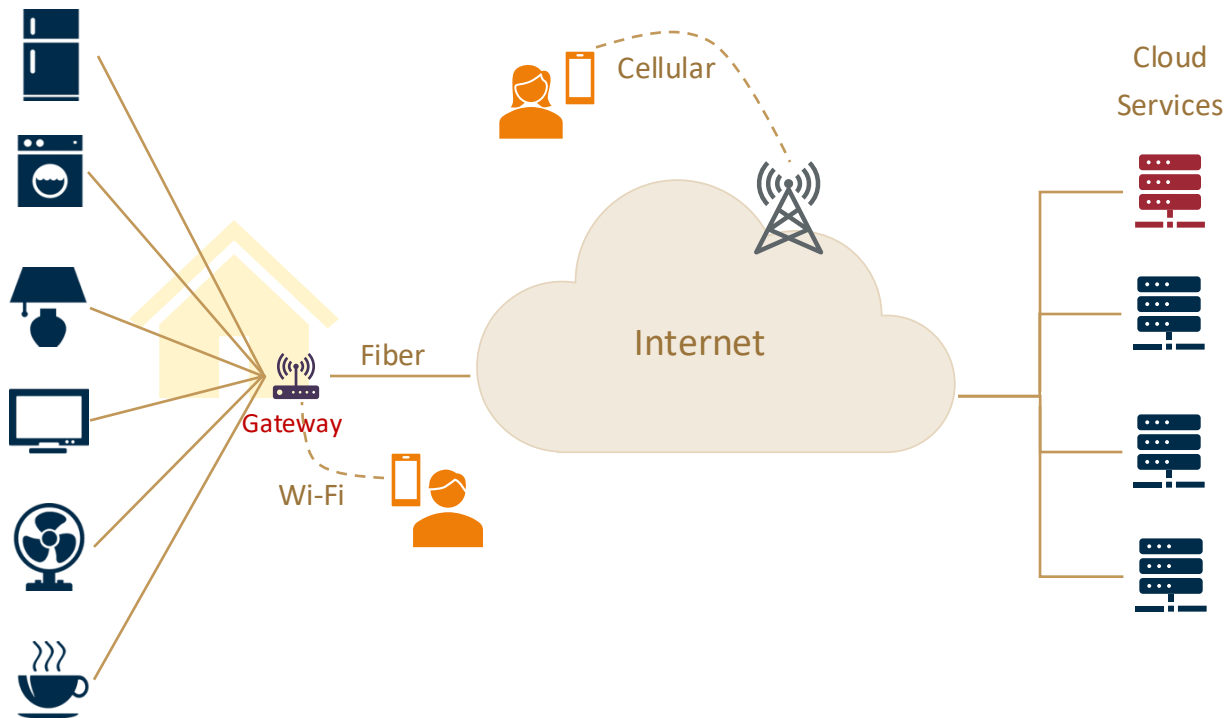
Security (for e.g., data encryption, firewalling)



Scalability problem: as the number of devices grows, so will the number of gateways that are required

Home automation

Home appliances



- Gateway performs **protocol translation**
- Incorporates basic network intrusion detection system
- Cloud services continue to perform analytics

Cloud vs. Fog vs. Edge



Edge computing



Pushing compute power, communication capabilities, intelligence down at device level



Processing as much as possible where data is collected

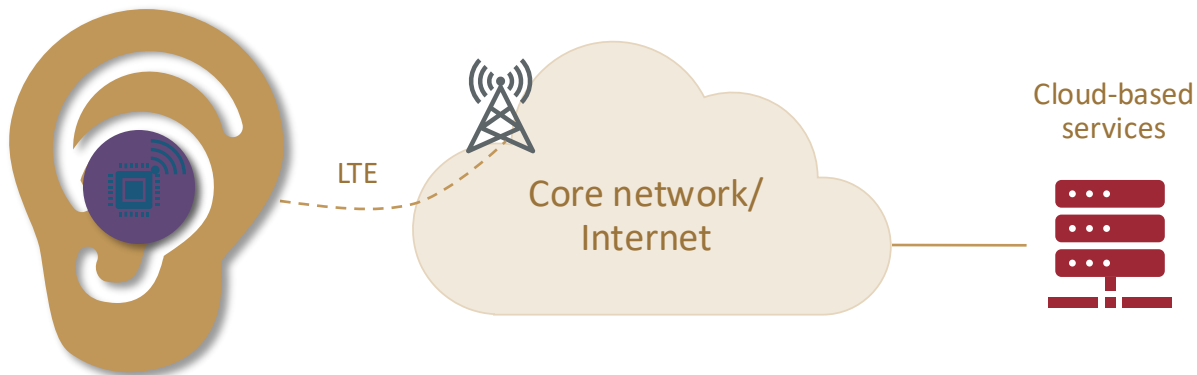


Transmitting only key information or summaries



Enabling new applications: automotive IoT, virtual/augmented reality, in-ear computing

Hearables

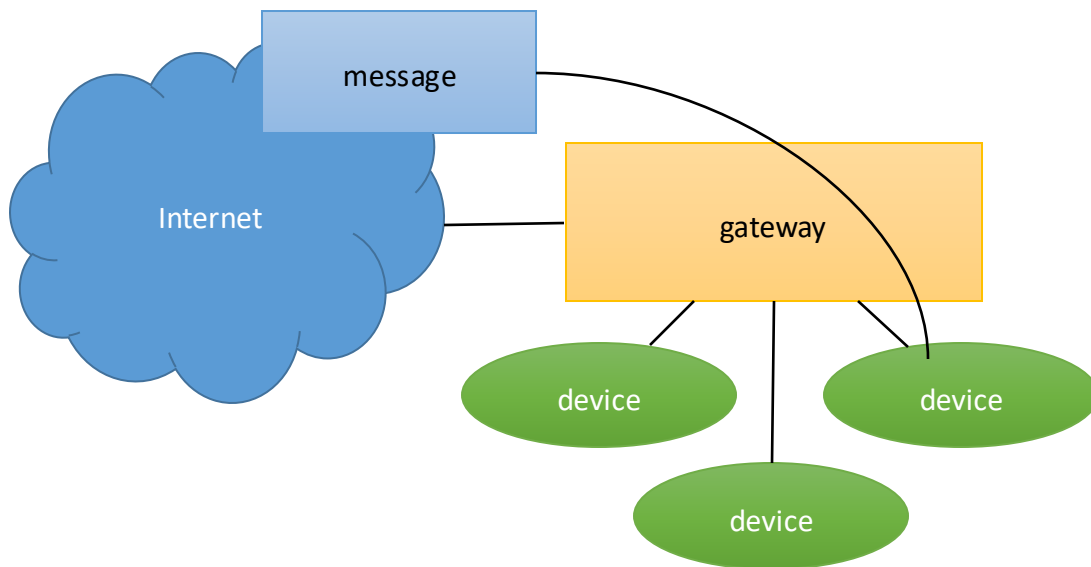


- **Hardware:** Low-power chips specialized in computationally intensive
- **Software:** **AI libraries** optimized for constrained devices

IoT Connectivity

IoT Networking Concepts

- **Ad hoc** network is self-organized---not set up by system administrator.
 - Edge device may not run IP protocol: IP connection may be provided by hub or gateway.



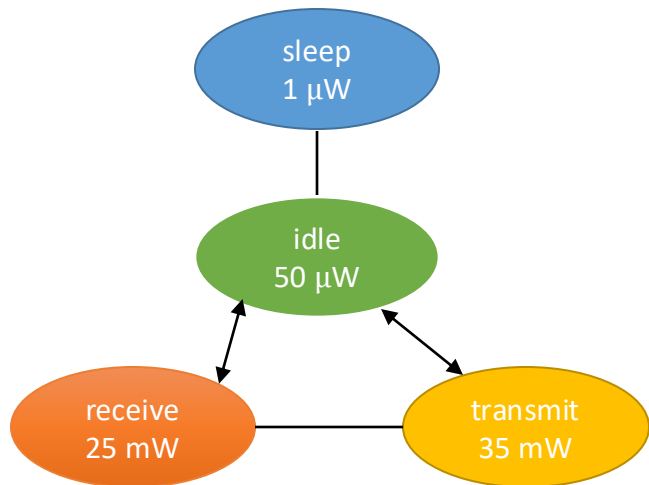
IoT Communication Technologies

- Signals collected by sensors can be transferred over ad hoc networks
- Connecting the different parts of networks to the sensors can be done by different technologies including
 - Wi-Fi, Bluetooth LE (Low Energy)
 - Low Power Wi-Fi
 - Ethernet
 - Long Term Evolution (LTE) (Cellular IoT)
 - Lora



Communications Energy

- Communications energy is a large part of node energy consumption.
- The transceiver (transmitter and receiver) might be put in different power modes to save energy.



step	state	time	energy
1	sleep	1 ms	1 nJ
2	idle	10 μs	0.5 nJ
3	receive	50 μs	1.25 nJ
4	transmit	50 μs	1.75 nJ
5	receive	50 μs	1.25 nJ
6	transmit	50 μs	1.75 nJ
			total = 7.5 nJ

Bluetooth technology

- Wireless technology working in the 2.4GHz band
 - 1Mb/s (Basic Rate – BR)
 - Communications range between ~0.5m and ~100m
- Enables a range of embedded devices to connect (and pair) with each other and exchange data (securely)
- Example applications: Wireless speakers, wearables, Internet-connected blood pressure monitors



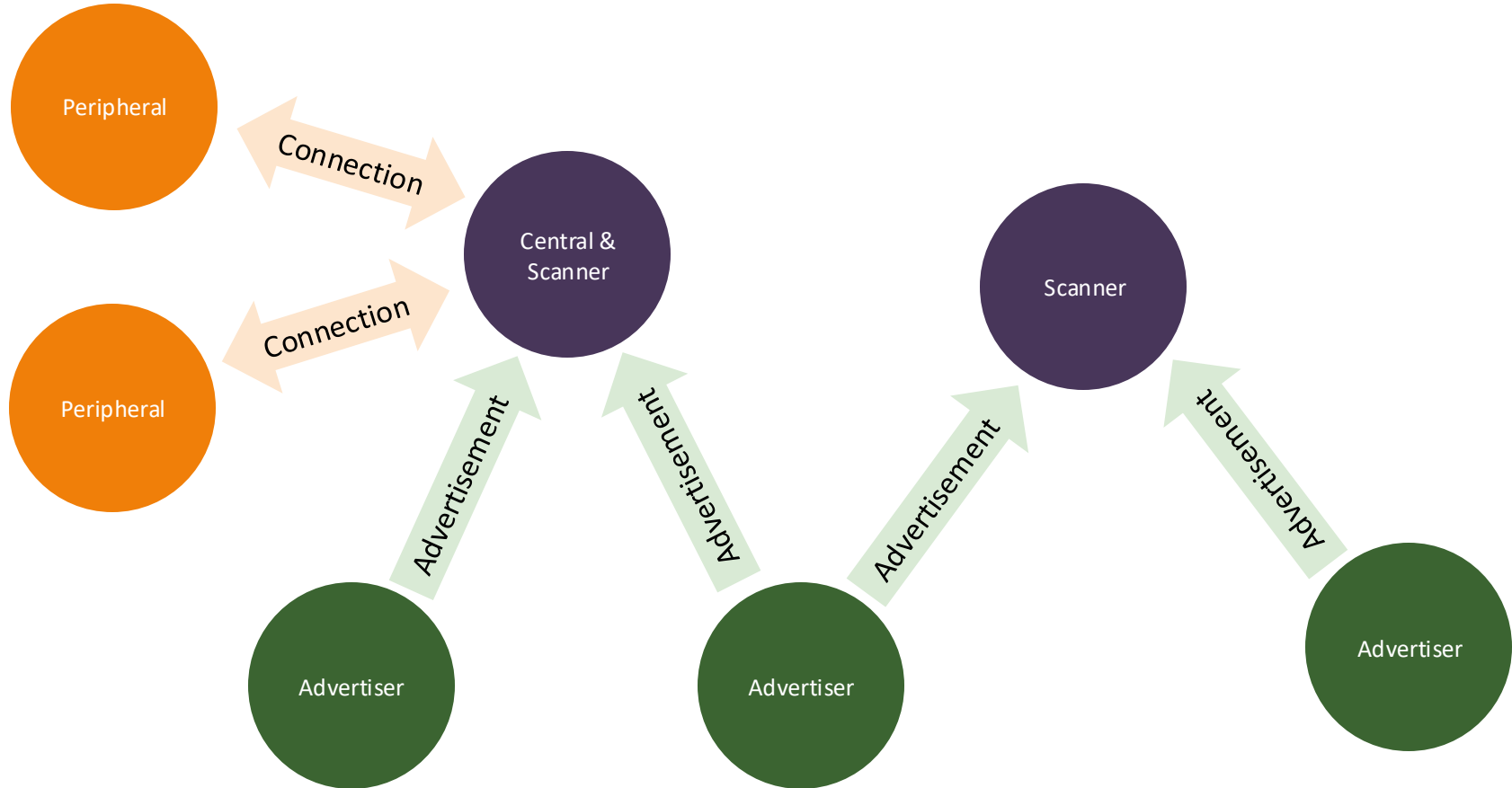
Basics of Bluetooth Low Energy (BLE)

- Direct device-to-device communication
 - Usually: Computer to Thing
 - Smartphone to device, Laptop to device, etc.
- Focus on making the “Thing” really low energy
 - Push energy-intensive requirements onto “Computer”

Two different communication modes in BLE

- Advertising
 - Discovery
 - Advertisements – broadcast messages indicating device details
 - Uni-directional communication from Advertiser to Scanner(s)
- Connections
 - Bi-directional communication between Peripheral and Central

BLE network topology



Bluetooth devices multiple roles at the same time

- A single device can have multiple roles simultaneously
 - Scanning and Advertising simultaneously
 - Peripheral and Scanner and Advertiser simultaneously
 - Peripheral and Scanner and Central and Advertiser simultaneously
- Also possible:
 - One Peripheral can be connected to multiple Central

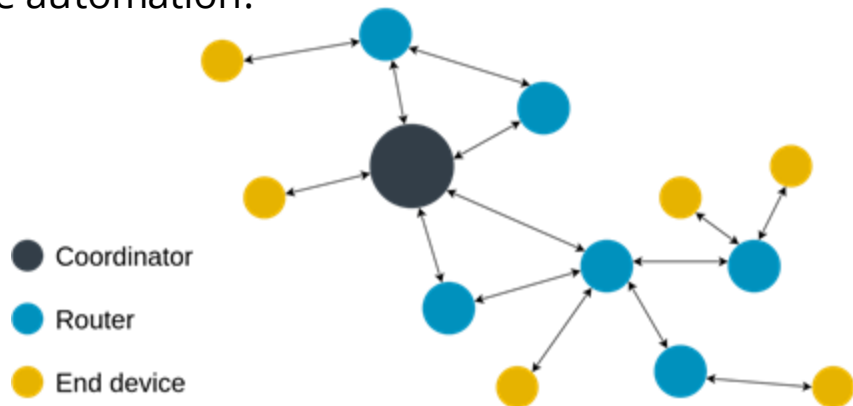
Bluetooth Low Energy (BLE)



- A number of revisions to the Bluetooth protocol to facilitate adoption on constrained low-powered devices (including IoT)
 - Faster procedure to set up a connection
 - Shorter transmission range (and therefore lower transmit power)
 - Radio interface more power efficient

ZigBee

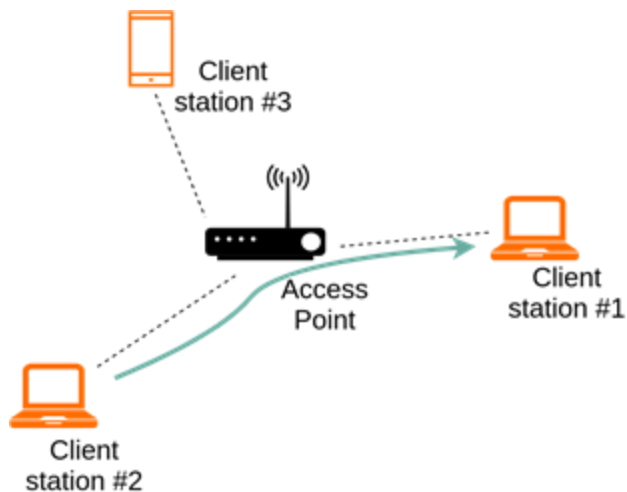
- Low-power low-rate personal area networking technology
 - Same low power characteristics of Bluetooth
 - Data rates: 20kb/s-250kb/s (lower than Bluetooth)
 - 10 to 100 m
- More suitable for applications where messages are less frequent and shorter;
 - For instance, home automation.



Wi-Fi



- Wireless networks following the **IEEE 802.11** specification are currently the **de facto WLAN technology**
 - Different protocols in the IEEE 802.11 family bundled together under the Wi-Fi trademark
- Two modes of operation: **infrastructure** and **ad-hoc**
 - Different nodes may play some roles of an AP in ad-hoc mode (e.g., periodic beaconing)

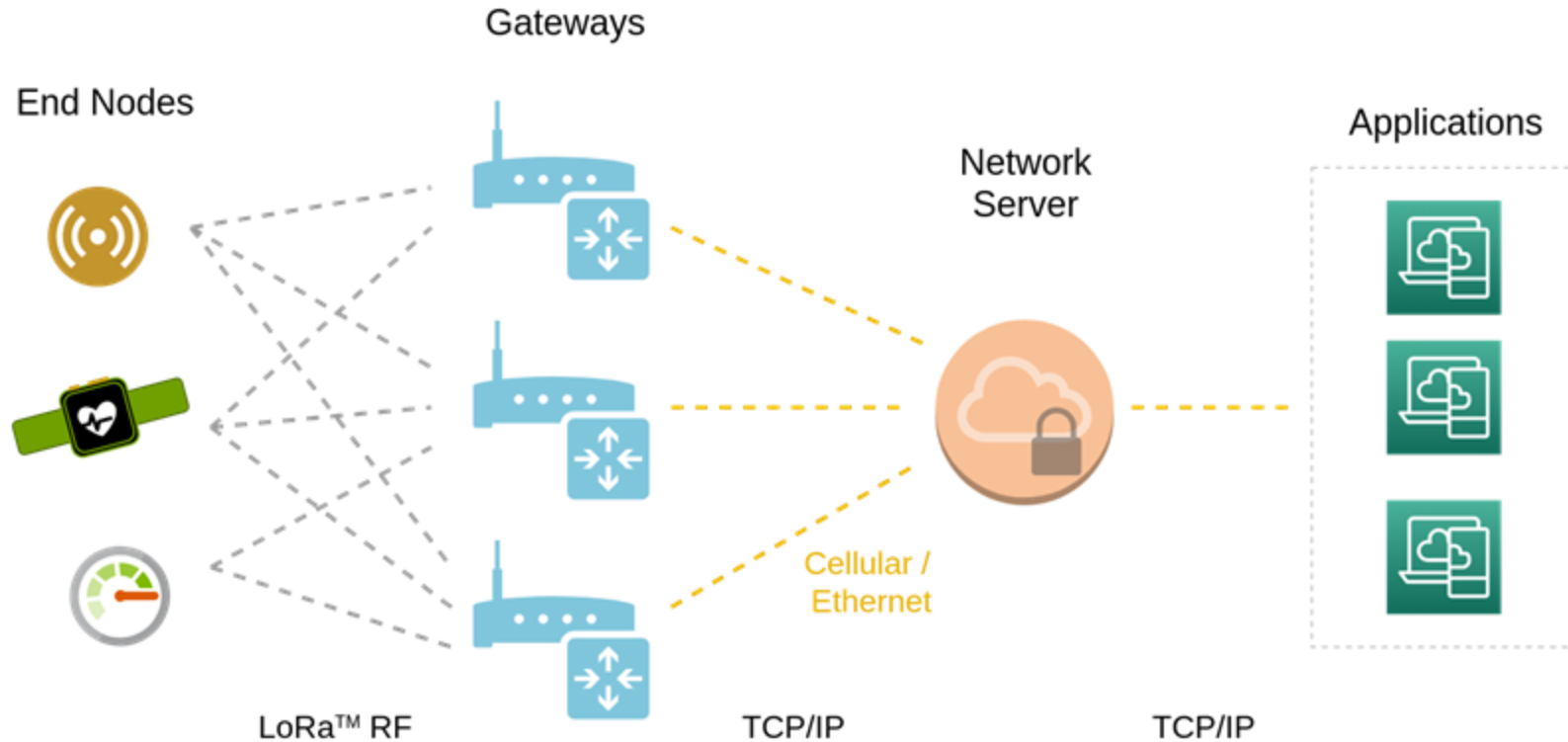


Long Range Wide Area Networking (LoRaWAN)

- Transmission ranges **over 10km** in rural areas, rates between 0.3 and 5.5kb/s



Long Range Wide Area Networking (LoRaWAN)



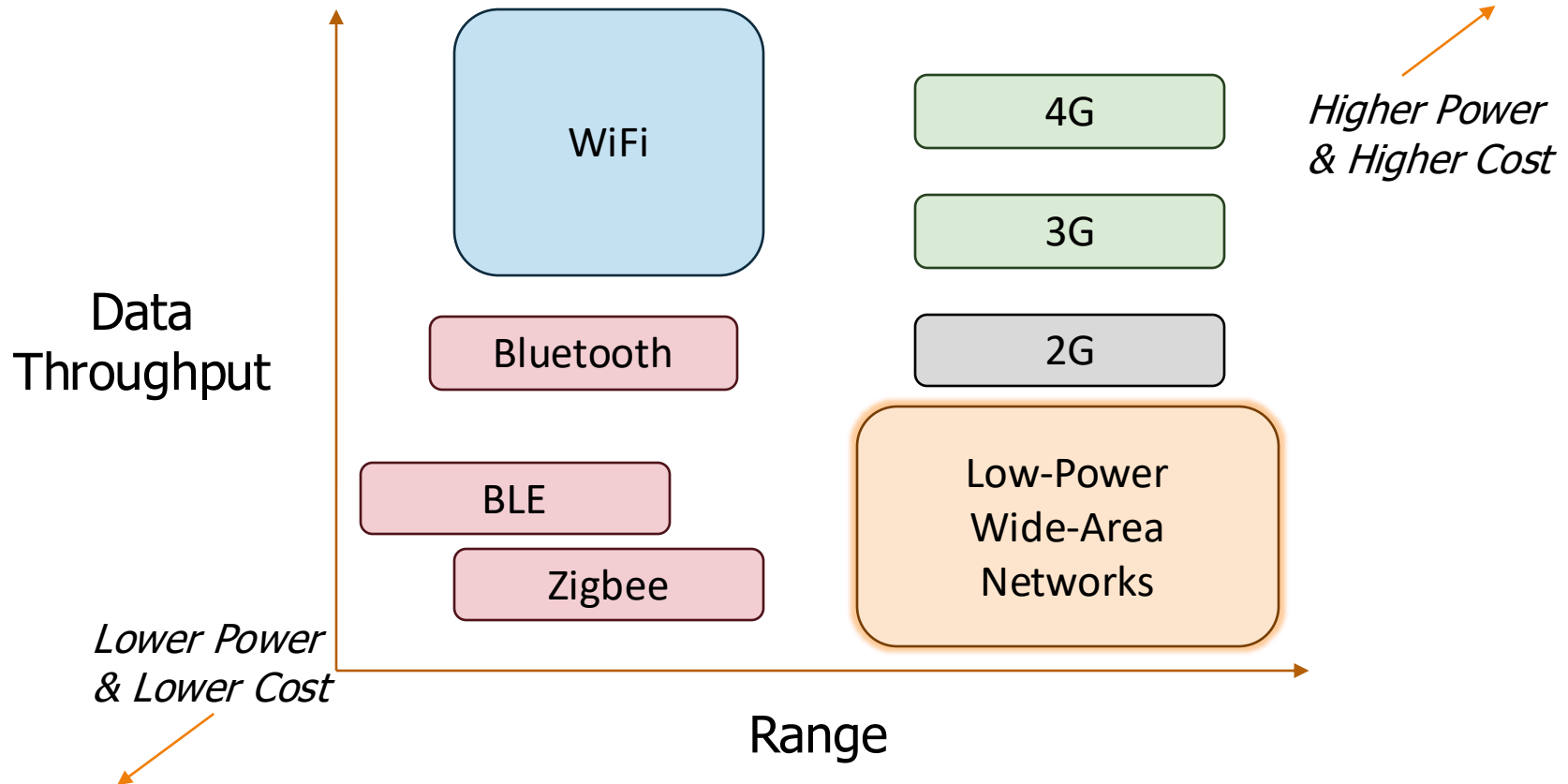
Narrow-band IoT (NB-IoT)

- Extension to cellular networks to support IoT applications
 - Particular focus on **reliability**
 - *licensed spectrum deterministic transmissions*
 - Applications: smart cities (fault diagnostics and control), eHealth
 - *requiring strict delay guarantees and reliability and where cost is not a critical issue.*
- Data rates: 25kb/s (downlink) and 64kb/s (uplink, multi-tone)
- Latency: < 10ms (transmissions aligned with LTE frames)

Which Technology?

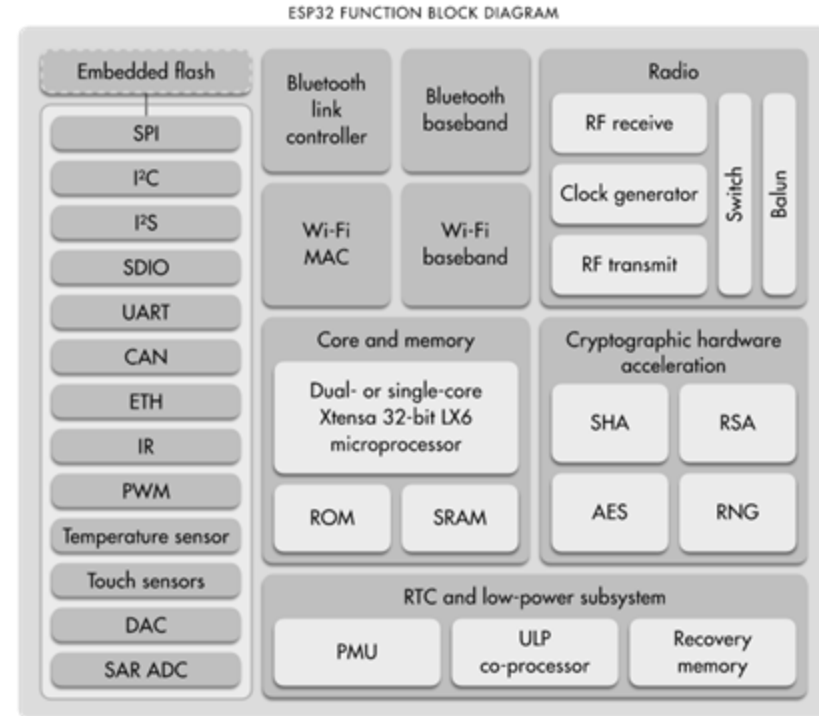
- Range
- Power Consumption
- Infrastructure
- Data rate

Long-range, low-data needs are not easily met



Wireless capability in microcontrollers

- ESP32
 - Microcontroller plus WiFi radio in single chip
 - *integrated Wi-Fi and Bluetooth connectivity*



Wireless Communication Primitives

- Send and Receive Interfaces

```
void sendData(void* data){  
    // Send over wifi (SPI/UART code)  
    ...;  
}
```

```
void receiveData(void* data){  
    // put into a circular buffer(SPI/UART code)  
    ...;  
}
```



Measure Temperature and Moisture

- I2C Communication

```
uint16_t measureTemp(void) {  
    tempConfig(); // configure the sensor  
    uint16_t temp = (uint16_t) tempDegC();  
  
    return temp;  
}
```

```
uint16_t measureMoist(void) {  
    moistConfig(); // configure the sensor  
    uint16_t moist = (uint16_t) moist();  
  
    return moist;  
}
```



Main Application

- Periodic Measurement Pseudo-code

```
void main(){
    ...
    setTimer(PERIOD);

    while(1){
        enterLowPowerMode();
        uint16_t temp = measureTemp();
        uint16_t moist = measureMoist();
        addToBuffer(temp, moist);
        calculateAverage();
        if(avgTemp > TEMPLIMIT || avgMoist > MOIST_LIMIT)
            sendData(ALARM);
    }
}

__interrupt_timer timer_isr(){
    wakeUpProcessor();
}
```

Cloud communication protocols

- Set up a lightweight HTTP service that follows the REpresentational State Transfer (REST) architectural style, i.e.,
 - Resources are accessible via Uniform Resource Identifiers (URIs)
 - *A set of standard methods is used to exchange representation of resources:*
 - e.g., POST, GET, etc.
 - Content can be accessed as HTML, XML, JSON, etc.
- Example:

```
record={  
    "date": "2020-02-28",  
    "time": "09:05:32",  
    "heart-rate": 67  
}
```

Cloud communication protocols

- Example (continued):
 - Representations sent to the server end-point using HTTP POST method
 - Python code snippet:

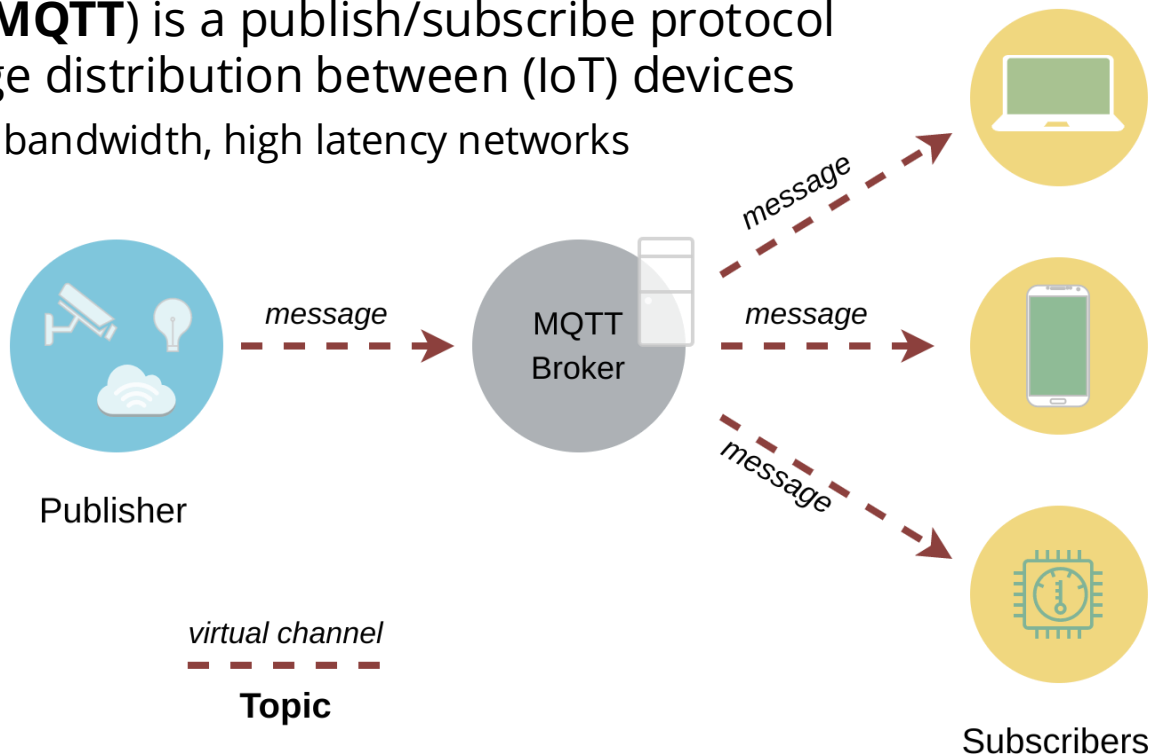
```
import requests  
import json
```

```
...
```

```
url = "http://<uri_of_server_end_point>"  
requests.post(url, data=json.dumps(activity))
```

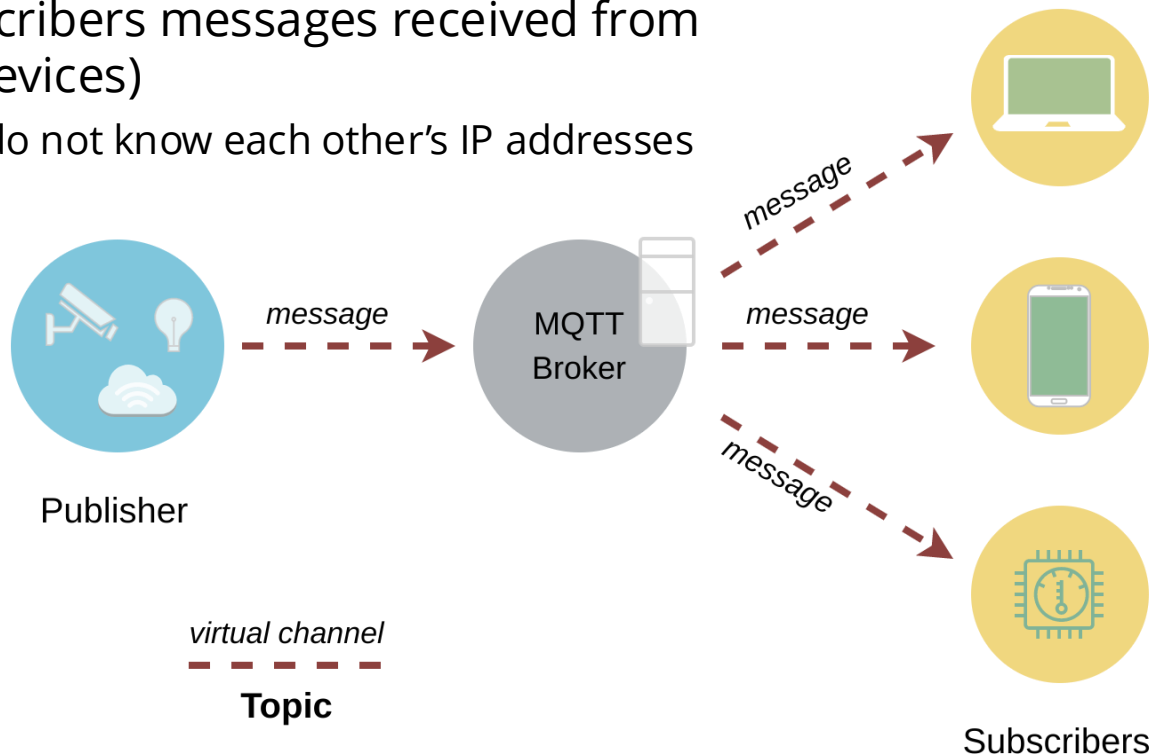
Cloud communication protocols

- MQ Telemetry Transport (**MQTT**) is a publish/subscribe protocol that enables short message distribution between (IoT) devices
 - Particularly suitable for low bandwidth, high latency networks



Cloud communication protocols

- Broker dispatches to subscribers messages received from publishers (different IoT devices)
 - Broker-centered → clients do not know each other's IP addresses



**How can you add wireless connectivity
to your projects?
Any ideas?**